

Automated and Portable Test Solutions for RF Devices

With recent developments in technologies like MIMO, femtocell and LTE, the RF device testing is reaching new dimensions. Read on to find out what's new in this field



Anritsu MS46524A with monitor

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Millions of devices these days operate and communicate in radio frequency (RF) range, which spans from 3kHz to 300GHz of the electromagnetic spectrum. From radio frequency identification (RFID) tags attached to clothes in shopping malls to Bluetooth headsets, and from Wi-Fi-enabled laptops to complex medical devices, this technology has become inevitable in our lives. This steep increase in the number of devices has led to the introduction of several industry standards. In order to bring out a new RF instrument to the market, the designer has to take it through a series of tests and ensure that the device meets the industry standards.

The testing and verification process for RF applications is done at two

stages—component level and system level. In component level testing and verification, each design component or building block of the system (amplifier, multiplexer, filter, modulator/demodulator, phase-locked loop, and transmitter and receiver antennae) is considered as a subsystem and its parameters (modulation scheme, frequency, sensitivity, saturation, etc) are tested independently. In system level verification, the tests and measurements are done on the RF device as a whole, and it simulates what exactly that system is going to perform.

With a wide range of applications and systems incorporating the RF principle and the extensive researches happening in this field, it is important for a design engineer to stay updated with the latest technologies and trends in RF communication.

Technology trends

The generally growing demand for higher throughput, faster results, customisability, reduced cost and better connectivity has led to greater researches in the field of wireless communication. Though the test and measurement field for RF devices also adapts to these enhancements, this field has not witnessed many path-breaking innovations in the past few years.

Analyses in real time. The previous generation of test and measurement equipment for RF devices faces a few challenges. With the dynamic nature of modern-day RF signals, it is necessary to understand the amplitude, frequency and modulation parameters over short and long intervals of time.

One of the most popular products these days is the real-time spectrum analyser that has an operating speed that matches the speed of the system it is simulating. In real-time spectrum analyser (RTSA) architecture, fast and continuous sampling and calculations are performed in order to keep up with the changing input signals. This helps the user to view events that were invisible in previous architectures like swept tuned analyser and vector signal analyser. Latest RTSAs let you trigger these events, selectively capture them into memory and analyse them extensively in multiple domains. R&S FSVR series by Rohde & Schwarz, N9030A and N9040A series by Keysight Technologies, and SPECMON RSA3000 and RSA5000 series by Tektronix are some examples of RTSA.

Heat dissipation. Proper power dissipation techniques are essential